Data Communication

Content 05: Digital Transmission

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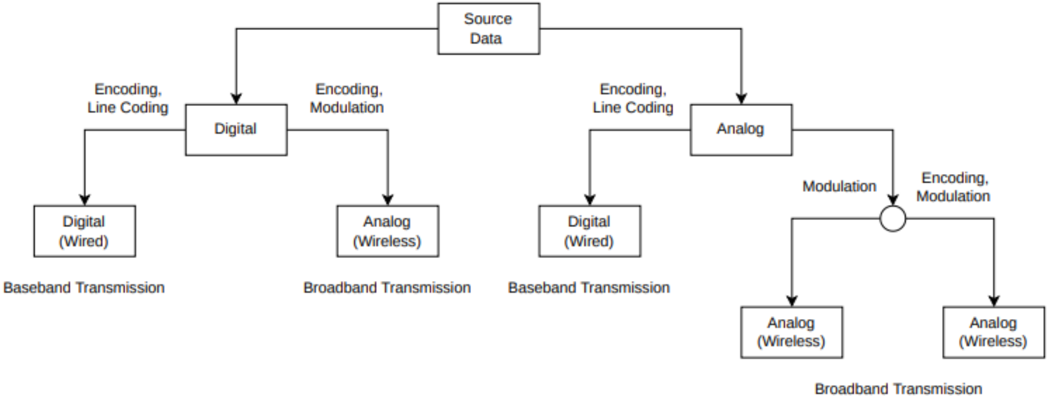
Data Transmission Pipeline



Baseband vs. Broadband Transmission

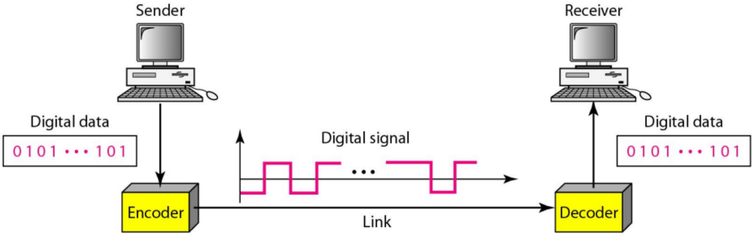
- **Baseband Transmission**
  - Transmits the original signal directly over the medium.
  - Does not require a carrier signal.
  - Entire channel bandwidth is used by a single signal.
  - Example: Ethernet (LAN).
- **Broadband Transmission**
  - Transmits the signal after modulating it onto a carrier signal.
  - Requires a high-frequency carrier signal.
  - Multiple signals can share the same medium using different carrier frequencies.
  - Example: Cable TV, Radio, Cellular communication.
- **Carrier Signal:** A high-frequency sinusoidal signal used to carry information by varying its amplitude, frequency, or phase (modulation).

What is Digital Transmission?

- Transmission of digital signals over a medium.
- Uses discrete values instead of continuous signals.
- Examples: Ethernet, Fiber-optic communication.

Digital-Digital Conversion

- Process of converting digital data into a digital signal.
- Uses line coding, block coding and scrambling techniques.
- Ensures proper transmission and synchronization.



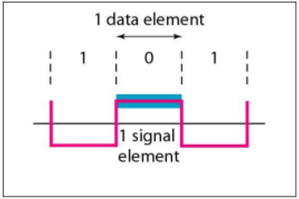
What is Line Coding?

- Encoding digital data into digital signals.
- Ensures compatibility with transmission medium.
- Different schemes affect bandwidth, synchronization, and error detection.

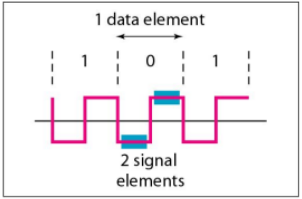
Signal Element and Data Element

- **Data Element**: Smallest unit of data (bit, character).
- **Signal Element**: Smallest unit of a signal that represents data.
- Relation: Data elements are mapped to signal elements for transmission.
- **Bit Rate**: Number of bits transmitted per second (bps).
- **Baud Rate**: Number of signal elements per second (baud).

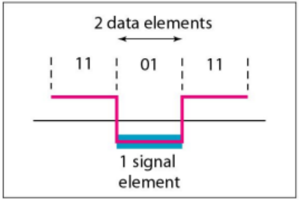
Signal Element and Data Element



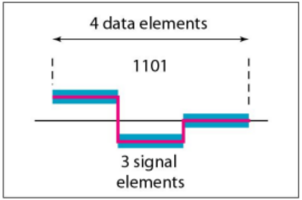
a. One data element per one signal element ($r = 1$)



b. One data element per two signal elements $\left(r = \frac{1}{2}\right)$



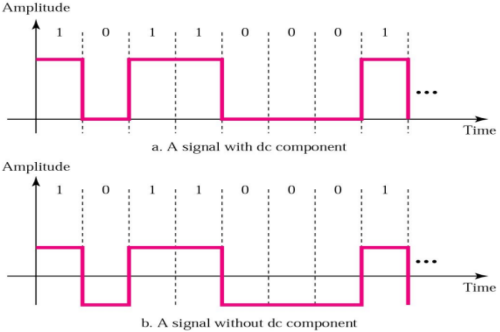
c. Two data elements per one signal element ($r = 2$)



d. Four data elements per three signal elements $\left(r = \frac{4}{3}\right)$

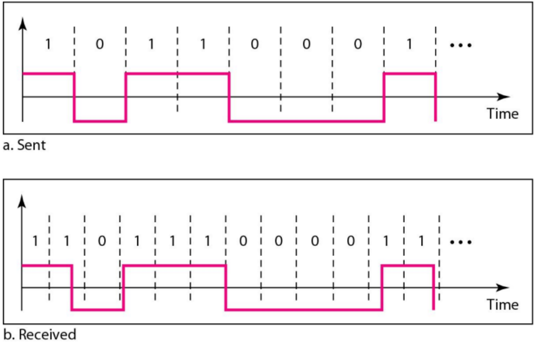
Problems of Line Coding: DC Component

- The DC component problem occurs when a signal has a constant voltage level, which can cause signal distortion or loss in certain transmission systems.
- DC component causes problems in AC-coupled transmission.
- Solution: Use bipolar encoding (e.g., AMI) to remove the DC component.

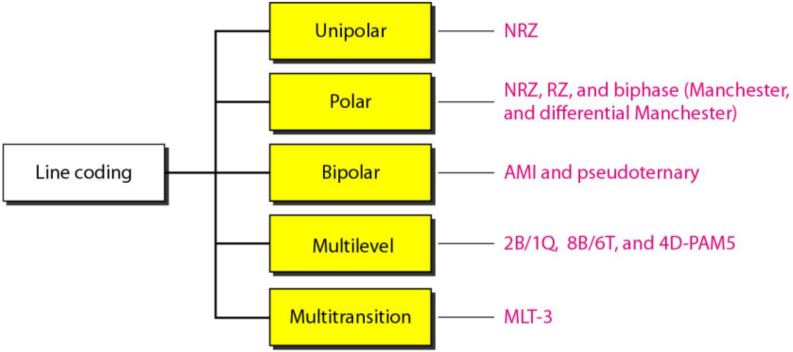


Problems of Line Coding: Self-Synchronization

- Long sequences of identical bits cause loss of synchronization.
- Self-clocking codes (e.g., Manchester encoding) solve this issue.

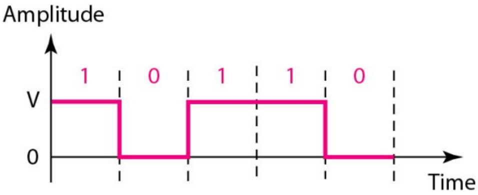


Classification of Line Coding



Unipolar NRZ

- One voltage level; 1=V, 0=0V
- Simple, easy to implement
- DC component, no synchronization



Polar NRZ-L and NRZ-I

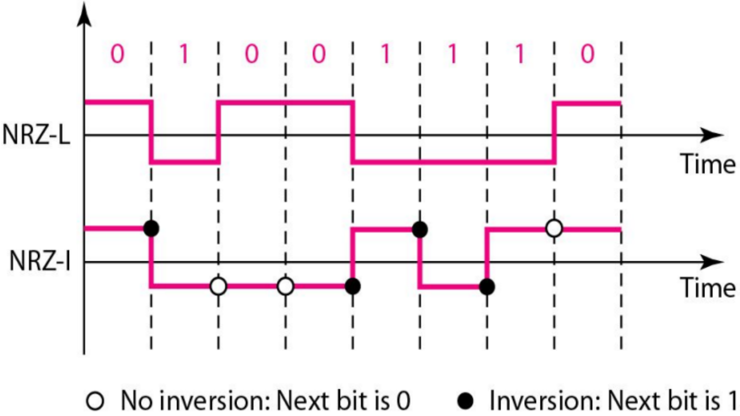
NRZ-L

- 1 = Negative, 0 = Positive voltage
- Reduces DC
- No synchronization

NRZ-I

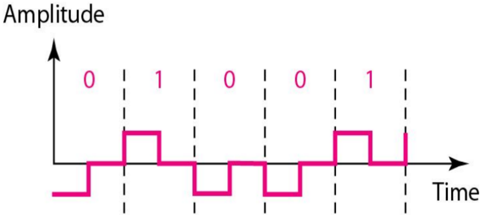
- Change at 1, No change at 0
- Better error detection
- Still no inherent synchronization

Polar NRZ-L and NRZ-I



Polar RZ

- 1 = Positive to Zero
- 0= Negative to Zero
- Self-clocking
- Requires more bandwidth



Manchester and Differential Manchester

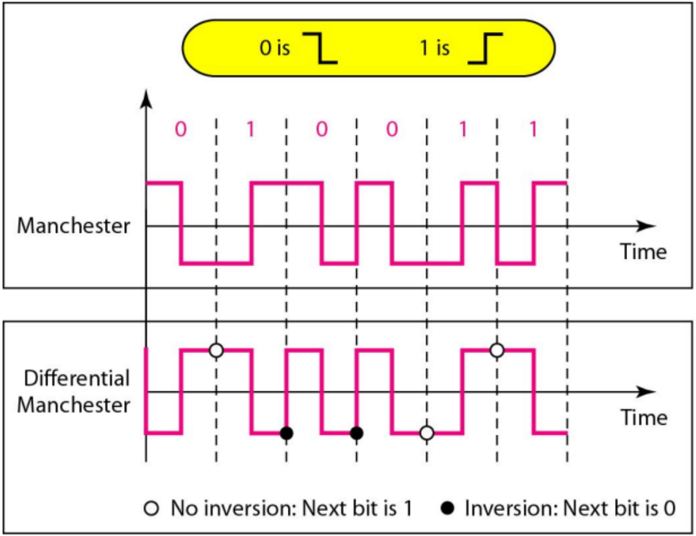
Manchester

- 1=Transition Low-High, 0=Transition High-Low
- Self-synchronizing
- Requires double bandwidth

Differential Manchester

- Transition at 0, No transition at 1
- Better noise immunity
- More complex decoding

Manchester and Differential Manchester



AMI and Pseudoternary

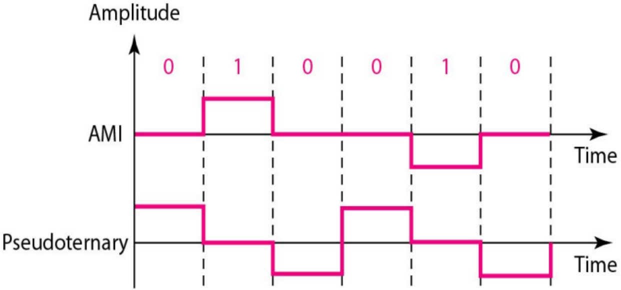
AMI

- 1 = Alternates between +V/-V
- 0 = 0 V
- No DC component
- Requires synchronization

Pseudoternary

- 0 = Alternates +V/-V, 1 = 0V
- No DC component
- Less common usage

AMI and Pseudoternary



Analog-Digital Conversion

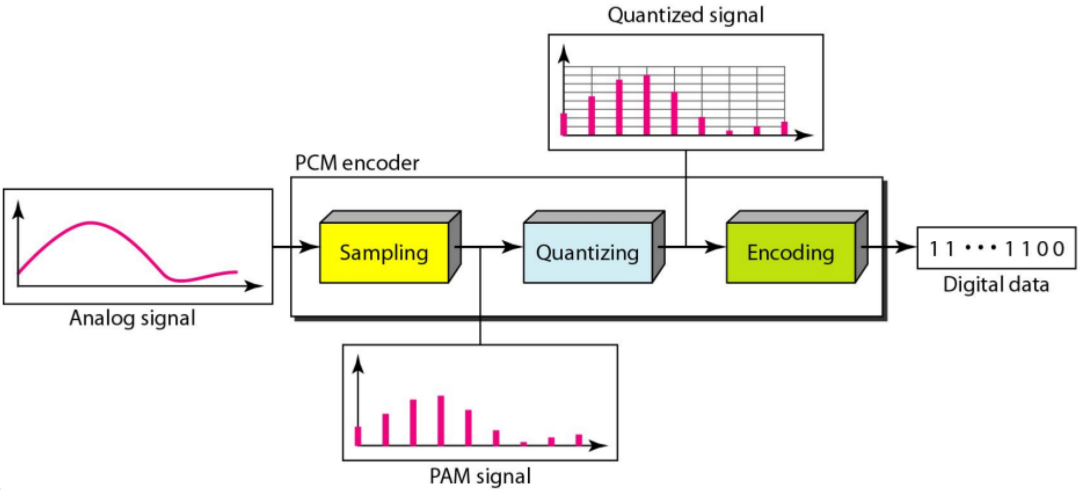
- Process of converting an analog signal into a digital signal for transmission.
- Enables reliable communication, storage, and digital signal processing.
- Common techniques:
  - Pulse Code Modulation (PCM)
    - Converts analog samples into binary-coded digital values.
    - Provides high signal quality and good noise immunity.
  - Delta Modulation (DM)
    - Encodes only the change (difference) between successive samples.
    - Simpler and requires less bandwidth than PCM, but may introduce distortion.

Pulse Code Modulation (PCM)

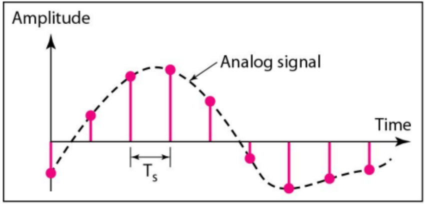
PCM is the most widely used technique for converting analog signals into digital form.

- It consists of three main steps:
    - Sampling – Measure the signal at regular time intervals.
    - Quantization – Approximate each sample to the nearest discrete level.
    - Encoding – Convert each quantized value into a binary code.

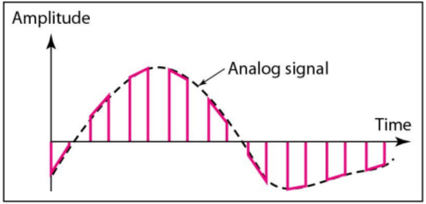
Pulse Code Modulation (PCM)



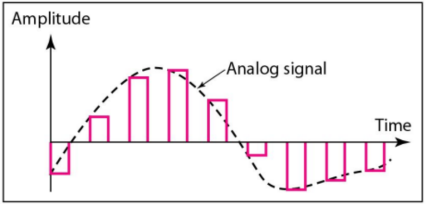
Pulse Code Modulation (PCM): Sampling



a. Ideal sampling



b. Natural sampling



c. Flat-top sampling

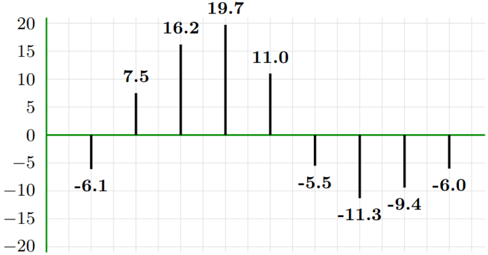
Pulse Code Modulation (PCM): Quantization & Encoding

Given extreme amplitudes: V_{max}, V_{min} and number of levels is L.

1. Calculate step size:
$$\Delta = \frac{V_{max} - V_{min}}{L}$$
2. Level number of a sample:
$$q = \frac{x_{sample} - V_{min}}{\Delta}$$
3. Convert level number q into an n-bit binary codeword.

Pulse Code Modulation (PCM): Worked Example

$V_{max} = +20V, V_{min} = -20V, L = 8$



Sample x	t_0	t_1	t_2	t_3	t_4	t_5	t_6	t_7	t_8
Amplitude (V)	-6.1	7.5	16.2	19.7	11.0	-5.5	-11.3	-9.4	-6.0
Level $q = \left\lfloor \frac{x - V_{min}}{\Delta} \right\rfloor$	2	5	7	7	6	2	1	2	2
Code	010	101	111	111	110	010	001	010	010

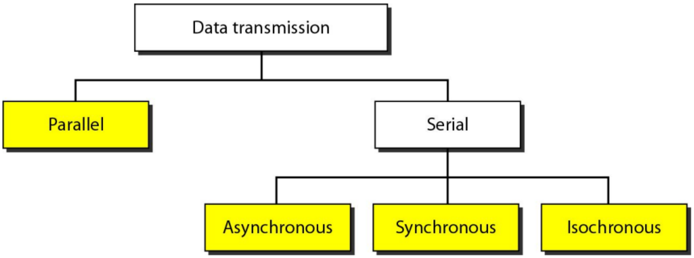
010 101 111 111 110 010 001 010 010

Transmission Modes

- Data transmission is the process of sending data from one device to another over a communication medium, such as wires or wireless signals.
- It can happen in different ways, like sending multiple bits at once (parallel) or one bit at a time (serial).

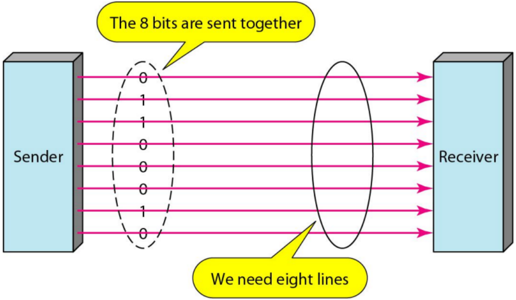
Classification of Transmission Modes

- Parallel Transmission - Multiple bits sent at a time.
- Serial Transmission - Bits sent one after another.



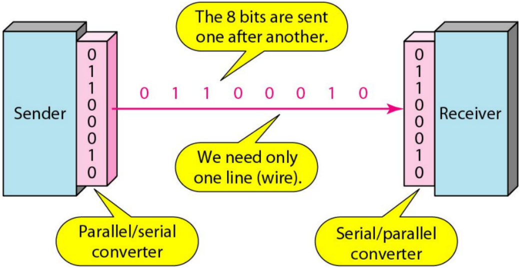
Parallel Transmission

- Data is sent multiple bits at a time using multiple channels.
- Faster than serial transmission but limited to short distances.
- Used in **computer buses, RAM, and printers**.



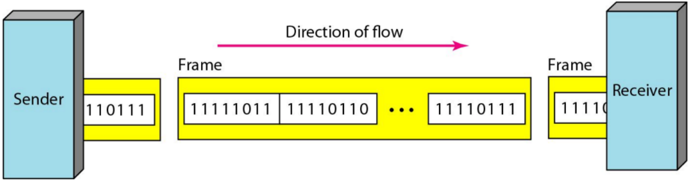
Serial Transmission

- Data is sent one bit at a time over a single channel.
- Used for long-distance communication (e.g., USB, Ethernet).
- Slower than parallel but requires fewer wires.



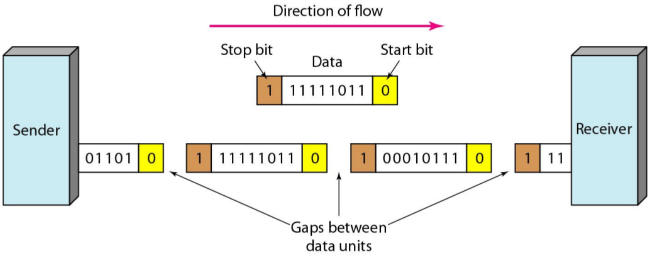
Synchronous Transmission

- Data is sent in a continuous stream with a clock signal.
- No start/stop bits; relies on synchronization between sender and receiver.
- Used in high-speed networks (Ethernet, SDH, ISDN, etc.).



Asynchronous Transmission

- Data is sent in small independent units (frames).
- Each unit has start and stop bits for synchronization.
- Slower than synchronous but cost-effective and simple.
- Used in keyboards, RS-232 serial communication.



Practice Problems

- Theoretical:
  Data Communication and Networking (4th Ed) - Behrouz A. Forouzan
  Chapter 04 - 2, 5, 11
- Mathematical:
  Data Communication and Networking (4th Ed) - Behrouz A. Forouzan
  Chapter 04 - 15, 16, 17, 18

References

- Data Communication and Networking (4th Ed) - Behrouz A. Forouzan
  Chapter 04 - 4.1, 4.3

Thank You